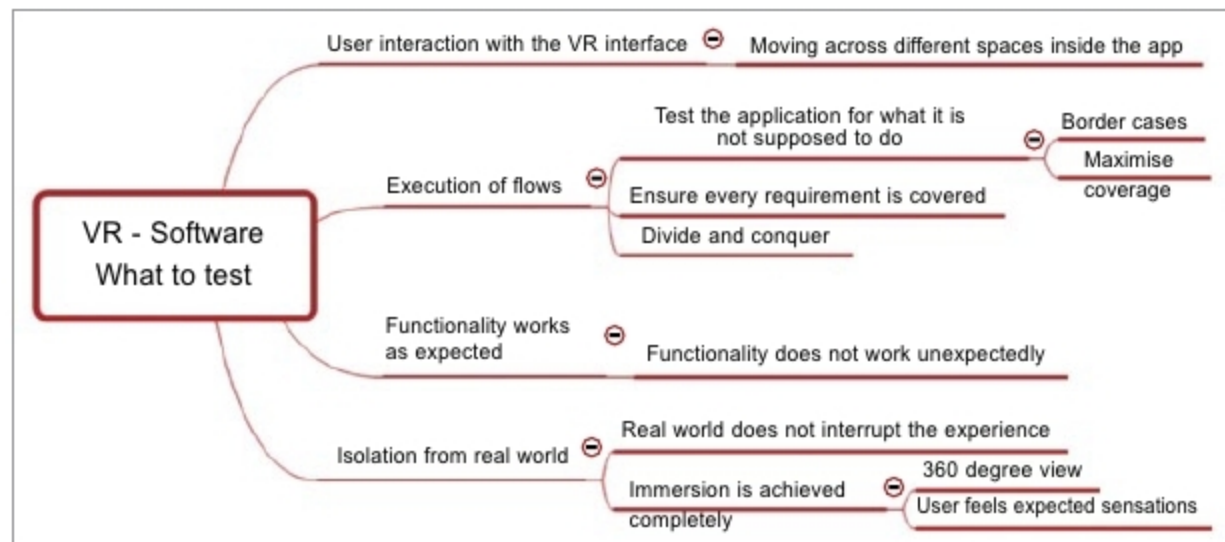




How To Check Performance Of VR Headsets

A good quality VR experience requires high consistent frame rate, low latency between head motion and display output, and frames that have to be pre-distorted to offset the effect of looking at the display through an optical lens



VR usability testing methodology (Credit: www.blazemeter.com)

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Virtual reality (VR) is described as a 3D, computer-generated environment that a person can explore and easily interact with. It is an artificial environment created and presented to the user in such a way that the user believes and accepts it as a real environment.

The technology utilises VR headsets or multi-projected environments to generate realistic sounds, images and other sensations that can simulate a person's actual physical presence within a virtual or imaginary setting. Being the future of technology, with escalation in the popularity of VR applications, demand for VR testing is also increasing.

VR and augmented reality (AR) often depend on specialised hardware. Proper functioning of these products can be tested by using the devices listed in the requirements, although VR and AR apps need more extensive and in-depth testing because these are more sophisticated software.

Objective and purpose of VR testing

Visual details and display performance are the biggest aspects that are monitored while testing a VR headset. Not only does the resolution shown to each eye matter, but refresh

rate, pixel structure and viewing angle can mean the difference between an immersive VR gaming experience or tunnel vision and motion sickness. The purpose of testing is to evaluate the overall impact of the environment on the device and how all this affects the user in the virtual environment.

Design and ergonomics are also important factors considered during testing. VR can affect different people in different ways, but a well-built design and an easy-to-adjust headband can make the experience enjoyable.

Hardware and software testing in VR applications is a challenge that needs innovative thinking, creative solutions, and a strong combination of technical skills and visual acuity. Firms need innovative approaches for manual and test automation to deliver quality consumer products, service and content in this space.

Diagnosing performance issues

In VR, if you are rendering frames at less than 60fps, some frames appear to render at 60fps and some at 30fps. This is because frames that miss Vsync interval have to wait for the next sync point. To optimise VR user experience, app rendering should be kept at 60fps or above. This helps maintain